Finding Matrix Inverses Using Elementary Row Operations (Gauss-Jordan Method)

High-Level Overview

In linear algebra, the **inverse** of a matrix acts like the reciprocal of a number. Just as multiplying a number by its reciprocal gives 1 ($5 \times \frac{1}{5} = 1$), multiplying a square matrix A by its inverse A^{-1} yields the **Identity Matrix** (I):

$$A \cdot A^{-1} = A^{-1} \cdot A = I$$

The **Gauss-Jordan Method** is a systematic algorithmic technique used to find this inverse using **elementary row operations**.

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1. The Gauss-Jordan Algorithm

To find the inverse of a square matrix A of order n:

1. **Set up an Augmented Matrix:** Construct a combined matrix $[A \mid I]$ by placing the given matrix A on the left and the identity matrix I of the same order on the right, separated by a vertical line.
2. **Apply Row Operations:** Perform a series of elementary row operations on the entire augmented matrix $[A \mid I]$ to transform the left side (A) into the identity matrix I.
3. **Extract the Inverse:** Once the left side successfully becomes I, the right side automatically transforms into the inverse matrix, A^{-1}.

$$\text{Augmented Form: } [A \mid I] \xrightarrow{\text{Row Operations}} [I \mid A^{-1}]$$

> **Important Rule (Singular Matrices):** If during your row reductions you encounter a complete row of zeros on the left side, it means matrix A is **singular** (its determinant is zero), and **it has no inverse**.

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2. Worked Examples

Problem 1: Finding the Inverse of a 3×3 Matrix

Find the inverse of matrix:

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 3 \end{bmatrix}$$

Solution:

Step 1: Set up the augmented matrix $[A \mid I]$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Step 2: Swap Row 1 and Row 2 ($R_1 \leftrightarrow R_2$) to get a non-zero element at the top-left

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 3 & 0 & 0 & 1 \end{array}\right]$$

Step 3: Eliminate the bottom-left entry ($R_3 \to R_3 - R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right]$$

Step 4: Clear above the second row ($R_1 \to R_1 - R_2$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & -1 & 1 \end{array}\right]$$

Step 5: Scale Row 3 to make the pivot 1 ($R_3 \to R_3/2$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1/2 & 1/2 \end{array}\right]$$

Step 6: Clear the remaining elements above the bottom-right pivot ($R_2 \to R_2 - R_3$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1/2 & -1/2 \\ 0 & 0 & 1 & 0 & -1/2 & 1/2 \end{array}\right]$$

Conclusion: The left side is now the identity matrix I. The right side gives us A^{-1}:

$$A^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ 1 & 1/2 & -1/2 \\ 0 & -1/2 & 1/2 \end{bmatrix}$$

Problem 2: Dealing with a Singular Matrix (No Inverse)

Find the inverse of matrix:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & -1 & 4 \\ 1 & -2 & 4 \end{bmatrix}$$

Solution:

Step 1: Set up the augmented matrix $[A \mid I]$

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 2 & -1 & 4 & 0 & 1 & 0 \\ 1 & -2 & 4 & 0 & 0 & 1 \end{array}\right]$$

Step 2: Apply row operations to eliminate entries below the first pivot ($R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -3 & 4 & -2 & 1 & 0 \\ 0 & -3 & 4 & -1 & 0 & 1 \end{array}\right]$$

Step 3: Simplify further ($R_3 \to R_3 - R_2$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & -3 & 4 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -1 & 1 \end{array}\right]$$

Conclusion: Notice that the third row on the left-hand side has turned completely into zeros `[0 0 0]`. According to the fundamental rules of matrix inversion, this means A is a **singular matrix** and **has no inverse**.

Quick Exam Revision Summary

- **Goal:** Transform $[A \mid I]$ into $[I \mid A^{-1}]$ using elementary row operations.
- **Allowed Operations:**
    i. Interchanging two rows ($R_i \leftrightarrow R_j$).
    ii. Multiplying a row by a non-zero scalar ($R_i \to kR_i$).
    iii. Adding/subtracting a multiple of one row to another ($R_i \to R_i \pm kR_j$).
- **Red Flag:** If a row of all zeros appears on the left side during reduction, stop—the matrix is **singular** and **does not have an inverse**.